A Bookworm's Analysis of Reading-Centric Data

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SI 618

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CHAPTER 1

INTRODUCTION

So, Why Books?

- Around 11,000 books are published PER DAY
- It would take the average reader 61,800 years to read every book ever written[1]
- So...getting your book to the bestseller list is hard! Let's take a look at what makes a bestseller!

CHAPTER 2

DATA OVERVIEW

Audible Dataset

Audiobooks

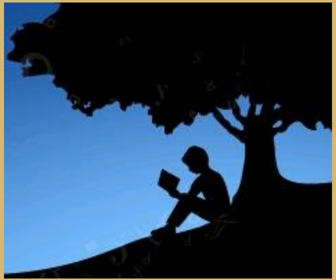
This dataset contains 6,368 entries. The data consisted of author name, book rating, number of reviews and book price.

Goodreads Dataset

Book Reviews

This dataset contains 11,123 entries containing information on book title, author, average rating, isbn, language, number of pages, number of ratings, reviews, publication date, and publisher.

Kindle Dataset



Digital Books

This dataset contains 133,102 entries. It holds information on book title, author, who sells it, star ratings reviews, price, if it is a Kindle Unlimited book, category, and more.

Preprocessing Plan

Address Null Values → Merge DataFrames → Find Weighted Average Rating → Add Author Dataset → Remove Columns

Address Null Values: We dropped rows that contained no reviews data.

Merge DataFrames: We used an <u>inner</u> merge to make sure that all rows had all information.

Find Weighted Average Rating: We used weighted averages to account for differences in the amount of ratings per DataFrame.

Add Author Dataset: We added an author dataset also found on Kaggle to supplement with author data.

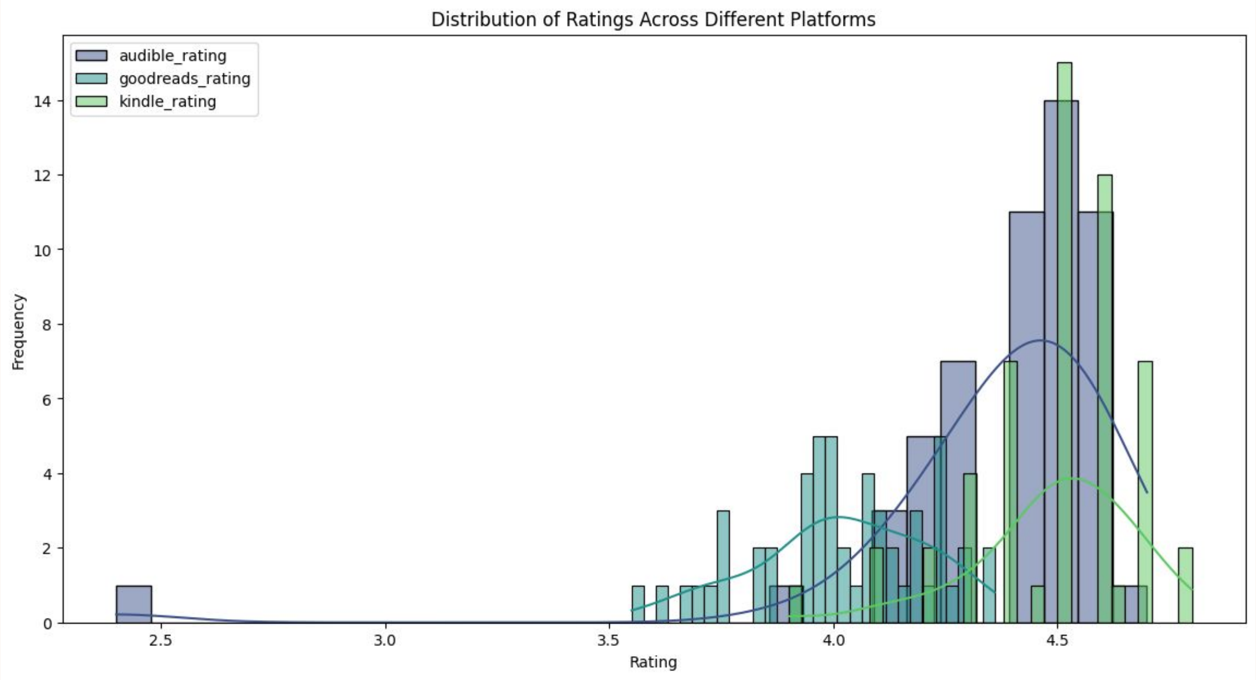
Remove Columns: We then removed the columns that were not to be used going forward.

CHAPTER 3

Data Analysis & Exploration

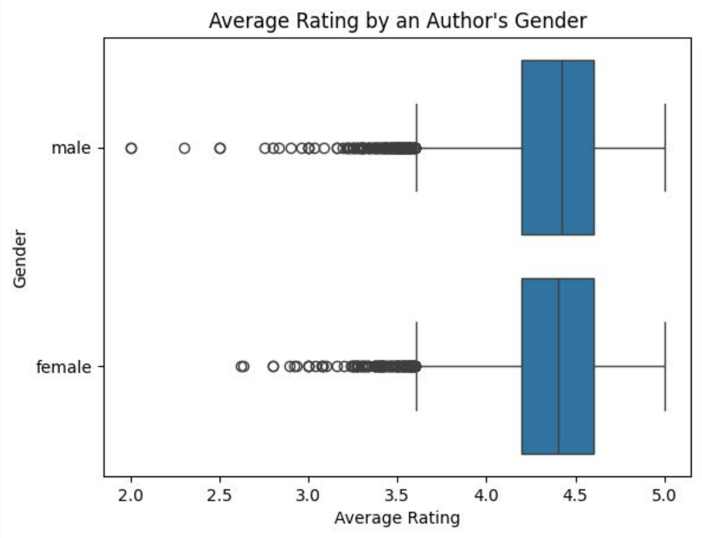
Analysis Results Pt. 1

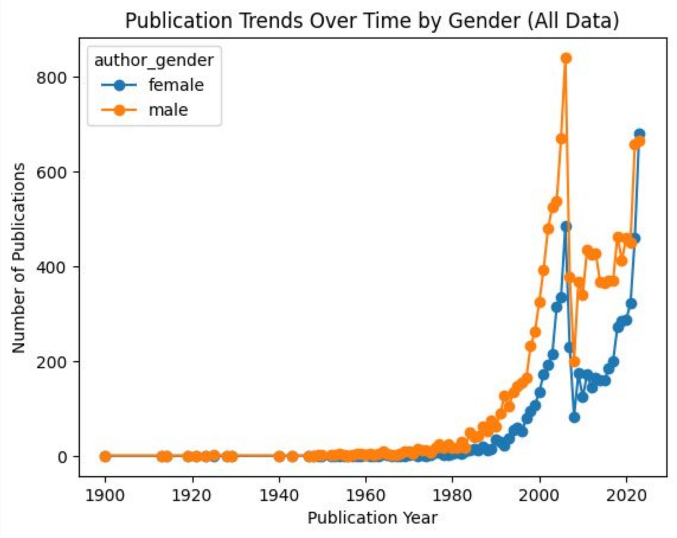
Overall, the ratings across the different data sets are skewed towards a higher rating but audible had a small peak at a lower rating.



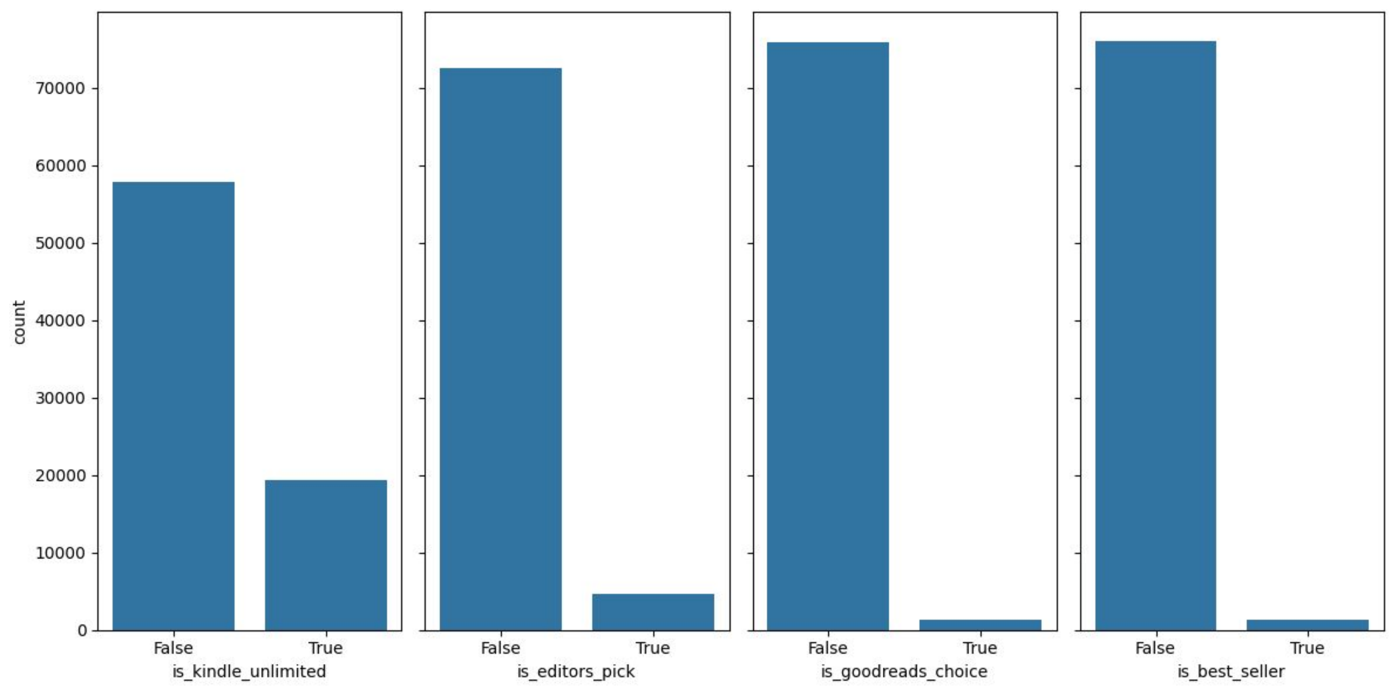
Analysis Results Pt. 2

Overall, there are more male authors than female authors and they follow a similar publishing trend until after 2020 when female authors increase sharply. We were not able to find a significant difference in ratings between the two genders.





Analysis Results Pt. 3



The “binary” categories that are True or False were generally unbalanced. This will be important later on for classifying this missing data.

CHAPTER 4

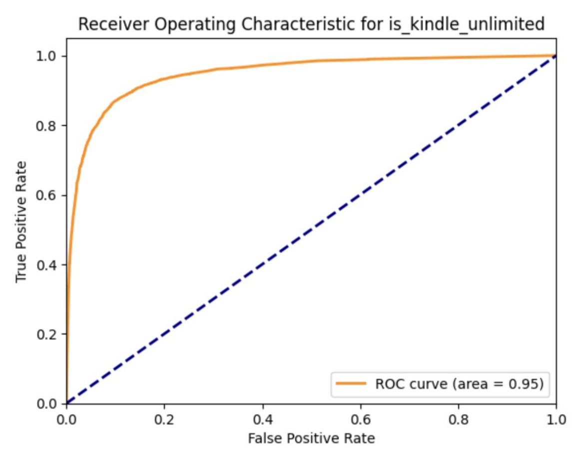
MACHINE LEARNING

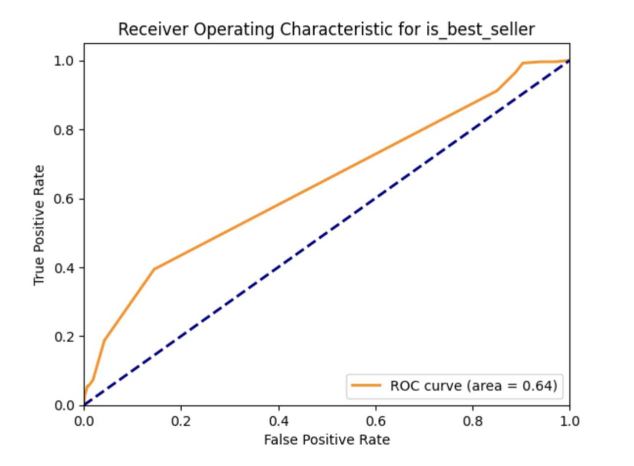
Classification

Given the nature and quality of our data, classification is the appropriate choice. It utilizes the labelled data that we have to build predictive models on four outcomes: if a book is on Kindle Unlimited, an editor's pick, Goodreads Choice, or a best seller. We used Random Forest and Decision Tree classifiers to fit our model and analyzed their accuracy.

Our model was the best at predicting if a book would be a Kindle Unlimited book with the Random Forest Classifier. It had a 90% accuracy and an ROC area of 95%.

The model with the lowest ROC area was the prediction for a book being a best seller with the Decision Tree classifier. This prediction only had an ROC area of 64%, even though the accuracy was at 98%.





CHAPTER 5

CONCLUSIONS & IMPLICATIONS

It is still very hard to determine if a book will be a bestseller!

Our model was falsely identifying books as not being best sellers, even if they were best sellers, resulting in skewed predictive data.

What is interesting is that the model did not predict any false positives, while having a substantial amount of false negatives. This could mean that when conflicted, the model defaulted to predicting a book to not being a best seller.

Book price and number of reviews were top influences on the model results, indicating that the model captures some relevant aspects of consumer behavior, but the high false-negative rate indicates potential gaps in how it interprets other factors or the overall distribution of data.

This unpredictability serves as a reminder that the magic of art lies in its ability to connect with people in ways that data and algorithms cannot fully anticipate. As much as we can measure trends, the spark that makes a story stick to a population of readers remains elusive and beautifully human.

References

“Reading Every Book.” Xkcd.com, 2022, what-if.xkcd.com/76/. Accessed 10 Dec. 2024.

All images of books in this presentation come from Amazon.com.

Thank You

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